PHYSICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

26. Three infinitely long wires with linear charge density λ are placed along the x-axis, y-axis and z-axis respectively. Which of the following denotes an equipotential surface?

(1) $xy + yz + zx$ = constant

(2) xyz = constant

(3) $(x^2 + y^2)(y^2 + z^2)(z^2 + x^2)$ = constant

(4) $(x + y)(y + z)(z + x)$ = constant

Answer (3)

Sol. Potential due to an infinite wire is $V = 2k\lambda \ln r$, where r is the distance from the wire.

Taking the point in space $P(x, y, z)$

Distance from wire along x-axis is $r_x = \sqrt{y^2 + z^2}$

Distance from wire along y-axis is $r_y = \sqrt{x^2 + z^2}$

Distance from wire along z-axis is $r_z = \sqrt{x^2 + y^2}$

$\Rightarrow$ Potential at P due to wire along x-axis is

$V_x = 2k\lambda \ln r_x$

Potential at P due to wire along y-axis is

$V_y = 2k\lambda \ln r_y$

Potential at P due to wire along z-axis is

$V_z = 2k\lambda \ln r_z$

$\Rightarrow$ Not potential at $P = V = V_x + V_y + V_z$

or $V = 2k\lambda \ln r_x + 2k\lambda \ln r_y + 2k\lambda \ln r_z$

i.e. $V = 2k\lambda \ln(r_x r_y r_z)$

or $V = 2k\lambda \ln\left(\sqrt{y^2 + z^2}\sqrt{z^2 + x^2}\sqrt{x^2 + y^2}\right)$

$= k\lambda \ln(y^2 + z^2)(z^2 + x^2)(x^2 + y^2)$

$\Rightarrow$ For equipotential surface

$(x^2 + y^2)(y^2 + z^2)(z^2 + x^2)$ = constant

27. Choose the correct nuclear process from the below options

[p: proton, n: neutron, e^-: electron, e^+: positron, ν : neutrino, $\bar{\nu}$: antineutrino]

(1) $n \to p + e^+ + \nu$ (2) $n \to p + e^+ + \bar{\nu}$

(3) $n \to p + e^- + \bar{\nu}$ (4) $n \to p + e^- + \nu$

Answer (3)

Sol. For all nuclear processes, charge must be conserved. Also, when a release of an electron (e^-) is always accompanied by a release of an antineutrino ($\bar{\nu}$)

Hence, $n \to p + e^- + \bar{\nu}$ is the correct answer.

28. Consider a long thin conducting wire carrying a uniform current I. A particle having mass "M" and charge "q" is released at a distance "a" from the wire with a speed v_0 along the direction of current in the wire. The particle gets attracted to the wire due to magnetic force. The particle turns round when it is at distance x from the wire. The value of x is

[μ_0 is vacuum permeability]

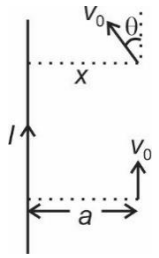
(1) $a\left[1 - \frac{mv_0}{q\mu_0 I}\right]$ (2) $\frac{a}{2}$

(3) $a\left[1 - \frac{mv_0}{2q\mu_0 I}\right]$ (4) $ae^{\frac{-4\pi mv_0}{q\mu_0 I}}$

Answer (4)



Sol. Let the velocity of the particle make an angle θ with initial direction when it is at a distance x.



$$\Rightarrow \quad \frac{dx}{dt} = -v_0 \sin\theta \qquad \ldots(i)$$

Also, $\dfrac{d\theta}{dt} = \dfrac{q}{m}\left(\dfrac{\mu_0 I}{2\pi x}\right)$ $\qquad \ldots(ii)$

$$\Rightarrow \quad \frac{dx}{d\theta} = \frac{-2\pi m v_0 x \sin\theta}{q\mu_0 I} \quad \ldots\text{dividing eq(i) with eq(ii)}$$

or $\displaystyle\int_a^x \frac{dx}{x} = \frac{-2\pi m v_0}{q\mu_0 I}\int_0^{\pi} \sin\theta\, d\theta$

$$\ln\frac{x}{a} = \frac{-4\pi m v_0}{q\mu_0 I}$$

Or $\quad x = ae^{\frac{-4\pi m v_0}{q\mu_0 I}}$

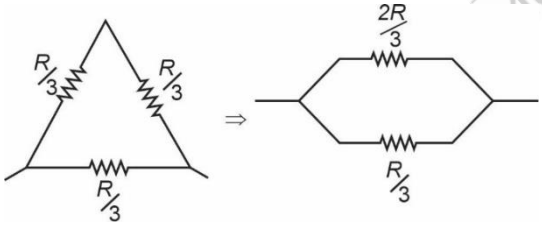
29. A wire of resistance R is bent into an equilateral triangle and an identical wire is bent into a square. The ratio of resistance between the two end points of an edge of the triangle to that of the square is

(1) 32/27 (2) 27/32

(3) 8/9 (4) 9/8

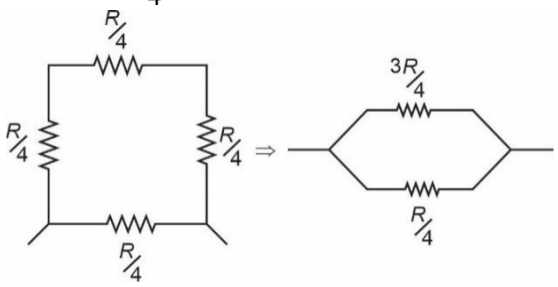
Answer (1)

Sol. For the wire bent into an equilateral triangle, each side has a resistance $\dfrac{R}{3}$.



$$R_{eq} = \frac{\left(\frac{2R}{3}\right)\left(\frac{R}{3}\right)}{\frac{2R}{3}+\frac{R}{3}} = \frac{2R}{9} = R_1 \text{ (lets say)}$$

For the wire bent into a square, each side has a resistance $\dfrac{R}{4}$.



$$R_{eq} = \frac{\left(\frac{3R}{4}\right)\left(\frac{R}{4}\right)}{\frac{3R}{4}+\frac{R}{4}} = \frac{3R}{16} = R_3 \text{ (lets say)}$$

$$\Rightarrow \quad \frac{R_1}{R_3} = \frac{\frac{2R}{9}}{\frac{3R}{16}} = \frac{32}{27}$$

30. A proton of mass 'm_p' has same energy as that of a photon of wavelength 'λ'. If the proton is moving at non-relativistic speed, then ratio of its de Broglie wavelength to the wavelength of photon is.

(1) $\dfrac{1}{2c}\sqrt{\dfrac{E}{m_p}}$ (2) $\dfrac{1}{c}\sqrt{\dfrac{E}{m_p}}$

(3) $\dfrac{1}{c}\sqrt{\dfrac{E}{2m_p}}$ (4) $\dfrac{1}{c}\sqrt{\dfrac{2E}{m_p}}$

Answer (3)

Sol. Energy of photon = $E = \dfrac{hc}{\lambda}$

$\Rightarrow$ Wavelength of photon = $\lambda = \dfrac{hc}{E}$

Energy of proton = $E = \dfrac{1}{2}m_p v^2 = \dfrac{P^2}{2m_p}$

$\Rightarrow$ Linear momentum of proton = $P = \sqrt{2m_p E}$

Or de-Broglie wavelength of proton

$$= \lambda_p = \frac{h}{P} = \frac{h}{\sqrt{2m_p E}}$$

Ratio $\dfrac{\lambda_p}{\lambda} = \dfrac{h}{\sqrt{2m_p E}} \times \dfrac{E}{hc}$

$$= \frac{1}{c}\sqrt{\frac{E}{2m_p}}$$



31. A Carnot engine (E) is working between two temperatures 473 K and 273 K. In a new system two engines – engine E_1 works between 473 K to 373 K and engine E_2 works between 373 K to 273 K. If η_{12}, η_1 and η_2 are the efficiencies of the engines E, E_1 and E_2 respectively, then

(1) $\eta_{12} = \eta_1 + \eta_2$ (2) $\eta_{12} \geq \eta_1 + \eta_2$

(3) $\eta_{12} < \eta_1 + \eta_2$ (4) $\eta_{12} = \eta_1 \eta_2$

Answer (3)

Sol. Efficiencies of a carnot engine $\eta = 1 - \frac{T_{sink}}{T_{source}}$

$$\Rightarrow \eta_1 = 1 - \frac{373\text{ K}}{473\text{ K}} = \frac{100}{473}$$

$$\eta_2 = 1 - \frac{273\text{ K}}{373\text{ K}} = \frac{100}{373}$$

$$\eta_{12} = 1 - \frac{273\text{ K}}{473\text{ K}} = \frac{100}{473}$$

$$\eta_{12} - \eta_1 = \frac{200}{473} - \frac{100}{473} = \frac{100}{473} < \frac{100}{373}$$

$$\Rightarrow \eta_{12} - \eta_1 < \eta_2$$

or $\eta_{12} < \eta_1 + \eta_2$

32. Due to presence of an em-wave whose electric component is given by $E = 100 \sin(\omega t - kx)$ NC^{-1}, a cylinder of length 200 cm holds certain amount of em-energy inside it. If another cylinder of same length but half diameter than previous one holds same amount of em-energy, the magnitude of the electric field of the corresponding em-wave should be modified as

(1) 50 $\sin(\omega t - kx)$ NC^{-1} (2) 400 $\sin(\omega t - kx)$ NC^{-1}

(3) 25 $\sin(\omega t - kx)$ NC^{-1} (4) 200 $\sin(\omega t - kx)$ NC^{-1}

Answer (4)

Sol. Energy density of an $EM_{wave} = \frac{1}{2}\varepsilon E_0^2$, whare E_0 is the amplitude of the wave.

Since total energy is same for both cylinders

$$\left(\frac{1}{2}\varepsilon E_1^2\right)\pi R_1^2 L_1 = \left(\frac{1}{2}\varepsilon E_2^2\right)\pi R_2^2 L_2$$

$$\Rightarrow E_1^2 R_1^2 L_1 = E_2^2 R_2^2 L_2$$

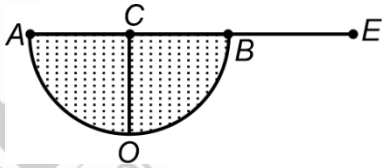
or $$E_2 = \frac{E_1 R_1}{R_2}\sqrt{\frac{L_1}{L_2}} = \frac{100\text{ d}}{(\text{d}/2)}\sqrt{\frac{L_1}{L1}} = 200\text{ N/C}$$

$$[\because L_1 = L_2 = 200\text{ cm}]$$

$\Rightarrow$ The amplitude of corresponding EM wave is 200 N/C

or the wave is $E = 200 \sin(\omega t - kx)$ NC^{-1}

33. A hemispherical vessel is completely filled with a liquid of refractive index μ. A small coin is kept at the lowest point (O) of the vessel as shown in figure. The minimum value of the refractive index of the liquid so that a person can see the coin from point E (at the level of the vessel) is _____.

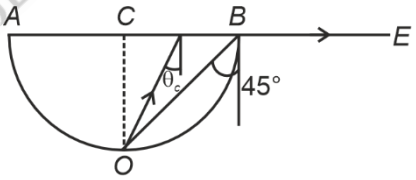


(1) $\sqrt{2}$ (2) $\sqrt{3}$

(3) $\frac{3}{2}$ (4) $\frac{\sqrt{3}}{2}$

Answer (1)

Sol.



For the rays from coin to reach the point E, the refracted rays must grazing the surface, *i.e.* they must be incident at critical angle θ_c inside the liquid.

$$\mu = \frac{1}{\sin\theta_c}$$

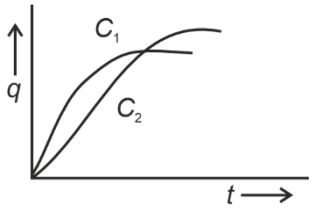
μ is minimum when θ_c is maximum.

Maximum value of $\theta_c = 45°$

$\Rightarrow$ μ has a minimum value of $\sqrt{2}$.



34. Two capacitors C_1 and C_2 are connected in parallel to a battery. Charge-time graph is shown below for the two capacitors. The energy stored with them are U_1 and U_2, respectively. Which of the given statements is true?



(1) $C_2 > C_1$, $U_2 < U_1$ (2) $C_1 > C_2$, $U_1 > U_2$

(3) $C_1 > C_2$, $U_1 < U_2$ (4) $C_2 > C_1$, $U_2 > U_1$

Answer (4)

Sol. For a capacitor at steady state

$q = CV$ and $U = \frac{1}{2}CV^2$

Since C_1 and C_2 are connected in parallel, $V_1 = V_2$.

Also from graph $q_1 < q_2$

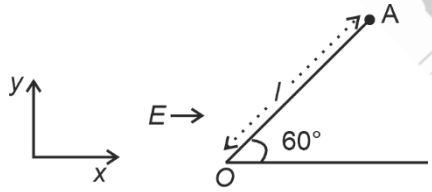
$\Rightarrow C_1V_1 < C_2V_2$

i.e. $C_1 < C_2$ or $C_2 > C_1$

$$\frac{U_1}{U_2} = \frac{C_1V_1^2}{C_2V_2^2} = \frac{C_1}{C_2} < 1$$

or $U_1 < U_2$ or $U_2 > U_1$

35. A particle of mass 'm' and charge 'q' is fastened to one end 'A' of a massless string having equilibrium length l, whose other end is fixed at point 'O'. The whole system is placed on a frictionless horizontal plane and is initially at rest. If uniform electric field is switched on along the direction as shown in figure, then the speed of the particle when it crosses the x-axis is



(1) $\sqrt{\frac{qEl}{m}}$ (2) $\sqrt{\frac{qEl}{4m}}$

(3) $\sqrt{\frac{qEl}{2m}}$ (4) $\sqrt{\frac{2qEl}{2m}}$

Answer (1)

Sol. From work energy theorem,

Work done by electric force = change in kinetic energy

or $qE\left(l - \frac{l}{2}\cos 60°\right) = \frac{1}{2}mv^2$

$\Rightarrow \frac{qEl}{2} = \frac{1}{2}mv^2$

or $v = \sqrt{\frac{qEl}{m}}$

36. In the experiment for measurement of viscosity 'η' of given liquid with a ball having radius R, consider following statements.

A. Graph between terminal velocity V and R will be a parabola.

B. The terminal velocities of different diameter balls are constant for a given liquid.

C. Measurement of terminal velocity is dependent on the temperature.

D. This experiment can be utilized to assess the density of a given liquid.

E. If balls are dropped with some initial speed, the value of η will change.

Choose the correct answer from the options given below:

(1) B, D and E only (2) A, C and D only

(3) A, B and E only (4) C, D and E only

Answer (2)

Sol. The terminal velocity of ball of radius R inside a liquid of viscosity η can be written as $V_T = \frac{2R^2g}{n}(\sigma - \rho)$, where σ is the density of ball and δ is the density of the liquid.

Hence,

A is correct since $V_T \alpha R^2$ gives a parabola on a graph

C is correct since $V_T \alpha \frac{1}{\eta}$ and η varies with temperature

D is correct since $V_T \alpha (\sigma - \rho)$ *i.e.*, varies with density of liquid.



37 Consider following statements:

A. Surface tension arises due to extra energy of the molecules at the interior as compared to the molecules at the surface, of a liquid.

B. As the temperature of liquid rises, the coefficient of viscosity increases.

C. As the temperature of gas increases, the coefficient of viscosity increases.

D. The onset of turbulence is determined by Reynold's number.

E. In a steady flow two stream lines never intersect.

Choose the correct answer from the options given below:

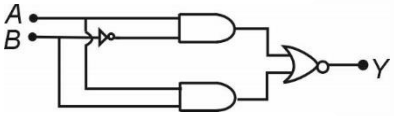
(1) A, D, E only (2) C, D, E only

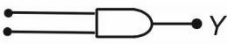
(3) A, B, C only (4) B, C, D only

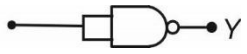
Answer (2)

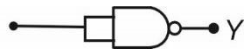
Sol. Surface tension arises due to extra energy of the molecules at the molecules at the surface as compared at the interior of a liquid. The coefficient of viscosity for a liquid decreases with rise in temperature where as it increases for gases with increase in temperature. The flow is turbulent for a Reynold's number greater than 2000. Stream lines never intersect in a steady flow.

38. Which of the following circuits has the same output as that of the given circuit?



(1) 

(2) 

(3) 

(4) 

Answer (3)

Sol. $Y = \overline{A\bar{B} + AB}$

$= \overline{A\left(\bar{B} + B\right)}$

$= \bar{A}$ as $\left(\bar{B} + B\right) = 1$

i.e. NOT gate with A as input

39. The centre of mass of a thin rectangular plate (fig – x) with sides of length a and b, whose mass per unit area (σ) varies as $\sigma = \frac{\sigma_0 x}{ab}$ (where σ_0 is a constant), would be _____.

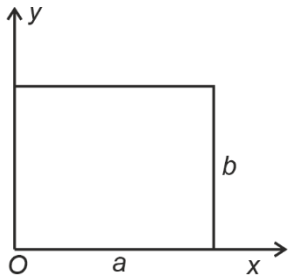


fig - x

(1) $\left(\frac{a}{2}, \frac{b}{2}\right)$ (2) $\left(\frac{1}{3}a, \frac{b}{2}\right)$

(3) $\left(\frac{2}{3}a, \frac{2}{3}b\right)$ (4) $\left(\frac{2}{3}a, \frac{b}{2}\right)$

Answer (4)

Sol. $dm = \sigma dA$

$= \sigma(dx)(dy) = \frac{\sigma_0 x}{ab}(dx)(dy)$

$$x_{com} = \frac{\int x dm}{\int dm} = \frac{\int x\frac{(\sigma_0 x)}{ab}(dx)(dy)}{\int \frac{\sigma_0 x}{ab}(dx)(dy)}$$

$$= \frac{\int_0^a x^2 dx \int_0^b dy}{\int_0^a x dx \int_0^b dy} = \frac{2a}{3}$$

$$y_{com} = \frac{\int y dm}{\int dm} = \frac{\int y\left(\frac{\sigma_0 x}{ab}\right)(dx)(dy)}{\int \frac{\sigma_0 x}{ab}(dx)(dy)}$$

$$= \frac{\int_0^a x dx \int_0^b y dy}{\int_0^a x dx \int_0^b dy} = \frac{b}{2}$$

i.e., $\vec{r}_{com} \equiv \left(\frac{2a}{3}, \frac{b}{2}\right)$



40. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**

Assertion A: A sound wave has higher speed in solids than gases.

Reason R: Gases have higher value of Bulk modulus than solids.

In the light of the above statements, choose the ***correct*** answer from the options given below

(1) **A** is true but **R** is false

(2) **A** is false but **R** is true

(3) Both **A** and **R** are true and **R** is the correct explanation of **A**

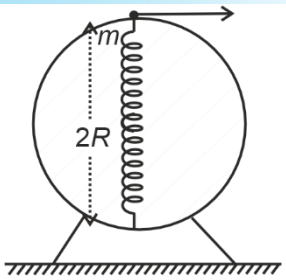
(4) Both **A** and **R** are true but **R** is **NOT** the correct explanation of **A**

Answer (1)

Sol. Speed of sound in a medium depends on inertial and elastic properties as $v=\sqrt{\frac{B}{P}}$ for gases and $v=\sqrt{\frac{Y}{P}}$ for solids. Since the elastic property of solid happens to be many folds greater than that of gases, the speed of sound in solids i5 higher than in gases.

Also, bulk modulus of gases varies between 0 and $\infty\left(B=-v\frac{dP}{dv}\right)$ hence reason is false.

41. A bead of mass 'm' slides without friction on the wall of a vertical circular hoop of radius 'R' as shown in figure. The bead moves under the combined action of gravity and a massless spring (k) attached to the bottom of the hoop. The equilibrium length of the spring is 'R'. If the bead is released from top of the hoop with (negligible) zero initial speed, velocity of bead, when the length of spring becomes 'R', would be (spring constant is 'k', g is acceleration due to gravity)

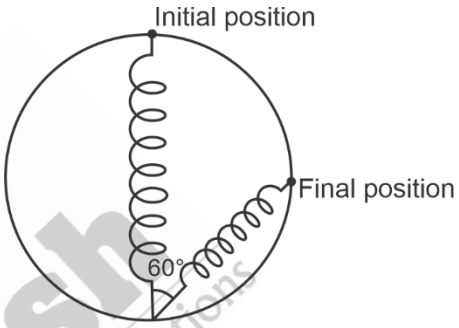


(1) $\sqrt{2Rg+\frac{4kR^2}{m}}$ (2) $\sqrt{2Rg+\frac{kR^2}{m}}$

(3) $\sqrt{3Rg+\frac{kR^2}{m}}$ (4) $2\sqrt{gR+\frac{kR^2}{m}}$

Answer (3)

Sol.



Work done by gravity $= mg(2R - R\cos 60°)$

$$= \frac{3mgR}{2}$$

Work done by spring $= -\frac{1}{2}k\left(0^2 - R^2\right)$

$$= \frac{1}{2}kR^2$$

Net work = change in kinetic energy

i.e. $\frac{3mgR}{2}+\frac{kR^2}{2}=\frac{1}{2}mv^2$

or $v^2=3gR+\frac{kR^2}{m}$

or $v=\sqrt{3gR+\frac{kR^2}{m}}$



42. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**

Assertion A : In a central force field, the work done is independent of the path chosen.

Reason R : Every force encountered in mechanics does not have not have an associated potential energy.

In the light of the above statements, choose the ***most appropriate*** answer from the options given below

(1) Both **A** and **R** are true but **R** is **NOT** the correct explanation of **A**

(2) **A** is true but **R** is false

(3) Both **A** and **R** are true but **R** is the correct explanation of **A**

(4) **A** is false but **R** is true

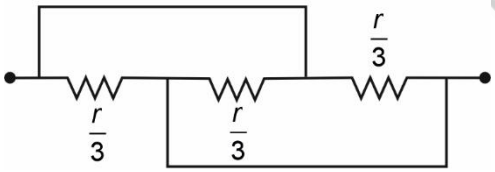
Answer (1)

Sol. Assertion is correct as central forces are conservative in nature, *i.e.* work done is independent of path.

Reason is true as some forces in mechanics like, friction are non-conservative because work done depends on path and only conservative forces have an associated potential energy.

Also, reason does not explain assertion.

43. Find the equivalent resistance between two ends of the following circuit



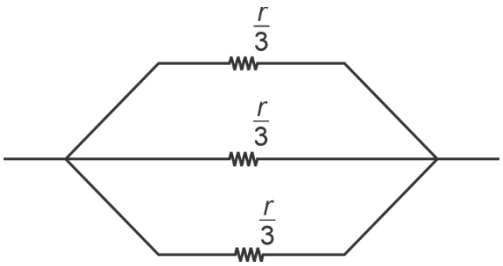
(1) $\frac{r}{6}$ (2) $\frac{r}{9}$

(3) r (4) $\frac{r}{3}$

Answer (2)

Sol. The three resistors are in parallel

i.e.



$$\Rightarrow \frac{1}{R_{eq}} = \frac{1}{\frac{r}{3}} + \frac{1}{\frac{r}{3}} + \frac{1}{\frac{r}{3}}$$

Or $R_{eq} = \frac{r}{9}$

44. A thin prism P_1 with angle 4° made of glass having refractive index 1.54, is combined with another thin prism P_2 made of glass having refractive index 1.72 to get dispersion without deviation. The angle of the prism P_2 is degrees is

(1) $\frac{16}{3}$

(2) 1.5

(3) 4

(4) 3

Answer (4)

Sol. For dispersion without deviation,

$$(\mu_1 - 1)A_1 = (\mu_2 - 1)A_2$$

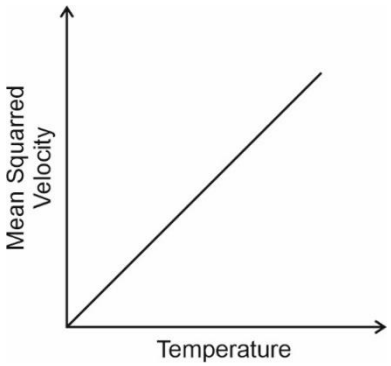
$$\Rightarrow (1.54 - 1)4° = (1.72 - 1)A_2$$

Or $A_2 = \frac{0.54}{0.72} \times 4 = 3°$

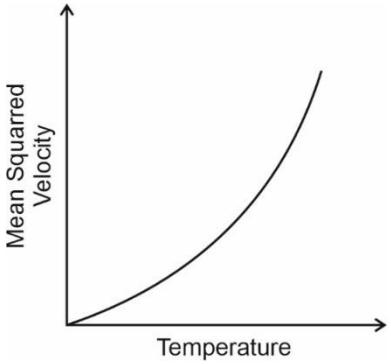


45. For a particular ideal gas which of the following graphs represents the variation of mean squared velocity of the gas molecules with temperature?

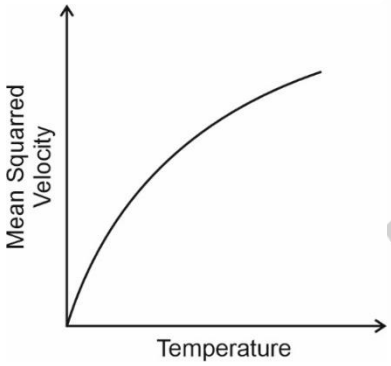
(1)



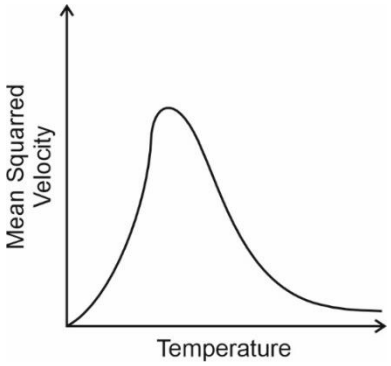
(2)



(3)



(4)



Answer (1)

Sol. Mean squared velocity $= \left(v_{rms}\right)^2 = \frac{3RT}{M}$

i.e. (mean squared velocity) $\propto$ (Temperature)

Hence, graph is a straight line.

SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

46. A double slit interference experiment performed with a light of wavelength 600 nm forms an interference fringe pattern on a screen with 10th bright fringe having its centre at a distance of 10 mm from the central maximum. Distance of the centre of the same 10th bright fringe from the central maximum when the source of light is replaced by another source of wavelength 660 nm would be _____ mm.

Answer (11)

Sol. Position of the nth bright fringe w.r.t. central maxima in a YDSE is $y_n = n\frac{\lambda D}{d}$.

$$\Rightarrow \quad \frac{y'_{10}}{y_{10}} = \frac{\lambda'}{\lambda}$$

or $$y'_{10} = \frac{\lambda'}{\lambda} y_{10}$$

$$= \left(\frac{660\text{ nm}}{600\text{ nm}}\right) 10\text{ mm}$$

$$= 11\text{ mm}$$



47. In a measurement, it is asked to find modulus of elasticity per unit torque applied on the system. The measured quantity has dimension of $[M^aL^bT^c]$. If $b = 3$, then value of c is _____.

Answer (0)

Sol. [Modulus of elasticity] = $ML^{-1}T^{-2}$

[Torque] = ML^2T^{-2}

[Modulus of elasticity per unit torque]

$$= \frac{ML^{-1}T^{-2}}{ML^2T^{-2}} = L^{-3}$$

48. Two iron solid discs of negligible thickness have radii R_1 and R_2 and moment of inertia I_1 and I_2, respectively. For $R_2 = 2R_1$, then ratio of I_1 and I_2 would be $1/x$, where x = _____.

Answer (16)

Sol. Given $\frac{M_1}{M_2} = \frac{\pi R_1^2}{\pi R_2^2} = \left(\frac{R_1}{R_2}\right)^2 = \frac{1}{4}$

$$\frac{I_1}{I_2} = \frac{M_1}{M_2} \times \left(\frac{R_1}{R_2}\right)^2 = \frac{1}{4} \times \frac{1}{4} = \frac{1}{16} = \frac{1}{x}$$

$x = 16$

49. A tiny metallic rectangular sheet has length and breadth of 5 mm and 2.5 mm, respectively. Using a specially designed screw gauge which has pitch of 0.75 mm and 15 divisions in the circular scale, you are asked to find the area of the sheet. In this measurement, the maximum fractional error will be $\frac{x}{100}$ where x is _____.

Answer (3)

Sol. Least count of screw gauge = $\frac{\text{Pitch}}{\text{Number of circular scale divisions}}$

$$\Rightarrow \text{LC} = \frac{0.75 \text{ mm}}{15} = 0.05 \text{ mm}$$

Length, $\ell = 5$ mm

Breadth, $b = 2.5$ mm

Area = ℓb

$\Rightarrow$ % error in area = (% error in length) + (% error in breadth) = $\frac{\Delta\ell}{\ell} \times 100 + \frac{\Delta b}{b} \times 100$

$$= \frac{0.05}{5} \times 100 + \frac{0.05}{2.5} \times 100$$

= 3%

50. The moment of inertia of a solid disc rotating along its diameter is 2.5 times higher than the moment of inertia of a ring rotating in similar way. The moment of inertia of a solid sphere which has same radius as the disc and rotating in similar way, is n times higher than the moment of inertia of the given ring. Here, n = _____.

Consider all the bodies have equal masses.

Answer (4)

Sol. Given,

$M_{disc} = M_{ring} = M_{sphere} = m$(let's say) and $R_{sphere} = R_{disc}$

$I_{disc} = 2.5\ I_{ring}$

$$\frac{mR_{disc}^2}{4} = 2.5\frac{mR_{ring}^2}{2}$$

$$\Rightarrow R_{disc}^2 = 5R_{ring}^2$$

$$I_{sphere} = nI_{ring}$$

$$\frac{2}{5}mR_{disc}^2 = n\frac{mR_{ring}^2}{2}$$

$n = 4$



CHEMISTRY

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

51. Match the **LIST-I** with **LIST-II**

	LIST-I (Redox Reaction)		**LIST-II (Type of Redox Reaction)**
A.	$CH_{4(g)} + 2O_{2(g)} \xrightarrow{\Delta} CO_{2(g)} + 2H_2O_{(l)}$	I.	Disproportionation reaction
B.	$2NaH_{(s)} \xrightarrow{\Delta} 2Na_{(s)} + H_{2(g)}$	II.	Combination reaction
C.	$V_2O_{5(s)} + 5Ca_{(s)} \xrightarrow{\Delta} 2V_{(s)} + 5CaO_{(s)}$	III.	Decomposition reaction
D.	$2H_2O_{2(aq)} \xrightarrow{\Delta} 2H_2O_{(l)} + O_{2(g)}$	IV.	Displacement reaction

Choose the ***correct*** answer from the options given below:

(1) A-II, B-III, C-I, D-IV (2) A-IV, B-I, C-II, D-III

(3) A-III, B-IV, C-I, D-II (4) A-II, B-III, C-IV, D-I

Answer (4)

Sol. (A) $CH_4(g) + 2O_2(g) \xrightarrow{\Delta} 2CO_2(g) + H_2O(l) \Rightarrow$ combination reaction

(B) $2NaH(s) \xrightarrow{\Delta} 2Na(s) + H_2(g) \Rightarrow$ Decomposition reaction

(C) $V_2O_5(s) + 5Ca(s) \xrightarrow{\Delta} 2V(s) + 5CaO(s)$ $\Rightarrow$ Displacement reaction

(D) $2H_2O_2$ (aq.) $\xrightarrow{\Delta} 2H_2O(l) + O_2(g)$ $\Rightarrow$ Disproportionation reaction

52. Given below are two statements:

Statement I: D-glucose pentaacetate reacts with 2, 4-dinitrophenylhydrazine

Statement II: Starch, on heating with concentrated sulfuric acid at 100°C and 2-3 atmosphere pressure produces glucose.

In the light of the above statements, choose the ***correct*** answer from the options given below

(1) Both Statement I and Statement II are true

(2) Statement I is true but Statement II is false

(3) Statement I is false but Statement II is true

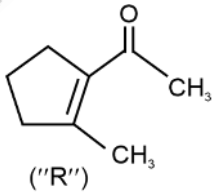
(4) Both Statement I and Statement II are false

Answer (3)

Sol. • D-glucose pentaacetate do not react with 2.4-DNP

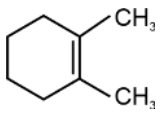
• Starch $\xrightarrow[\substack{2-3 \text{ atm} \\ 100°C}]{\text{conc.}H_2SO_4,\ \Delta}$ Glucose

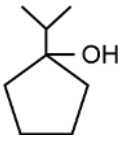
53. A molecule (''P'') on treatment with acid undergoes rearrangement and gives (''Q''). (''Q'') on ozonolysis followed by reflux under alkaline condition gives (''R''). The structure of (''R'') is given below.

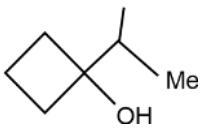


(''R'')

The structure of (''P'') is

(1) 

(2) 

(3) 

(4) 